Proposition I.4
The Side–Angle–Side Congruence Theorem

Formal Reconstruction — Technical Documentation

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1 Introduction

Euclid's fourth proposition is the first theorem on triangle congruence.

Unlike the preceding propositions, it does not describe a geometric construction. Instead, it establishes that two triangles are congruent whenever two corresponding sides and the included angle are equal.

Euclid's original proof relies on the method of superposition, one of the most discussed arguments in the *Elements*. Modern axiomatic systems generally avoid superposition as a primitive method of proof.

In the present reconstruction over Hilbert geometry, the proposition is obtained from the Side–Angle–Side (SAS) congruence theorem, which has already been established within the Hilbert theory.

Thus, while the mathematical content coincides with Euclid's Proposition I.4, the logical structure of the proof is fundamentally different.

2 The Classical Configuration

Unlike the first three propositions of Book I, Proposition I.4 is not a geometric construction. Instead, it establishes a criterion for the congruence of two triangles.

The proposition considers two triangles satisfying three geometric hypotheses:

- two pairs of corresponding sides are congruent;
- the included angles are congruent.

The objective is to prove that the triangles are congruent. As a consequence, the remaining corresponding side and the remaining corresponding angles are also congruent.

2.1 Final Configuration

The geometric configuration is shown in Figure 1.

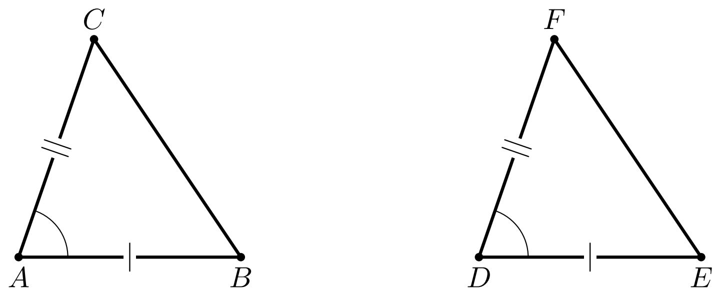

Figure 1: The SAS data of Proposition I.4: $AB \cong DE$, $AC \cong DF$, and $\angle BAC \cong \angle EDF$.

The hypotheses are

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF.$$

The desired conclusions are

$$BC \cong EF,$$

together with

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

3 Statement

Proposition I.4.

If two triangles have two corresponding sides congruent and the included angles congruent, then the triangles are congruent.

More precisely, let ABC and DEF be two non degenerate triangles:

$$AB \cong DE,$$

$$AC \cong DF,$$

and

$$\angle BAC \cong \angle EDF.$$

Then

$$BC \cong EF,$$

$$\angle ABC \cong \angle DEF,$$

and

$$\angle ACB \cong \angle DFE.$$

4 Classical Proof

Euclid's proof is a superposition argument. Conceive the triangle ABC as being applied to DEF so that A coincides with D and the ray AB coincides in direction with the ray DE. Since

$$AB \cong DE,$$

the point B must then coincide with E.

There is a further positional condition that is easy to miss in the bare classical diagram. The transported point C must be chosen on the same side of the line DE as F. With that choice, the equality

$$\angle BAC \cong \angle EDF$$

makes the ray AC coincide with the ray DF. Since also

$$AC \cong DF,$$

the point C coincides with F.

The endpoints of the remaining sides therefore coincide pairwise, so BC coincides with EF. The two triangles coincide under the superposition, and hence

$$BC \cong EF, \qquad \angle ABC \cong \angle DEF, \qquad \angle ACB \cong \angle DFE.$$

Euclid also concludes that the two triangles are equal in area. That additional magnitude statement belongs to the classical proposition but is not part of the present `TriangleCongruenceResult` interface.

□

Diagrammatic audit. Only one geometric figure is needed. Proposition I.4 has no auxiliary construction: the proof acts on the two triangles already present in the hypotheses. The figure therefore records exactly the SAS data and nothing more. Wilkins's discussion is useful here not because it supplies additional construction diagrams, but because it makes explicit the same-side condition on the transported third vertex that Euclid leaves implicit in words.

5 Proof Architecture of the Reconstruction

```
Euclid
------
AB ~= DE
AC ~= DF
angle BAC ~= angle EDF
        |
apply triangle ABC to triangle DEF
A -> D, ray AB -> ray DE
        |
B -> E
        |
choose the same side of DE for C and F
        |
equal included angle + AC ~= DF
        |
C -> F
        |
triangles coincide
        |
BC ~= EF and corresponding angles equal


Hilbert / Lean
--------------
two noncollinear triangles
+ two side congruences
+ included-angle congruence
        |
TriangleCongruentFromSAS
        |
TriangleCongruenceResult
        |
euclid_proposition_4
```

6 Hilbert Reconstruction

The foundational organization is different. Superposition is not replayed inside Proposition I.4. The required rigidity is already available through the Hilbert SAS interface.

For two noncollinear triangles ABC and DEF, the hypotheses

$$AB \cong DE, \qquad AC \cong DF, \qquad \angle BAC \cong \angle EDF$$

are precisely the inputs of `TriangleCongruentFromSAS`. Its result is a `TriangleCongruenceResult`, which packages the corresponding side and angle congruences.

Thus the difference from Euclid is foundational rather than mathematical: Euclid proves SAS by superposition; the Hilbert development has already established SAS before the Book I wrapper is reached.

7 Lean Formalization

The complete implementation is available in the repository. Here the entire mathematical step is a single interface call, so only that call is worth displaying:

```
exact
  TriangleCongruentFromSAS
    Geo
    A B C
    D E F
    hABC
    hDEF
    hAB
    hAngle
    hAC
```

The explicit hypotheses $hABC$ and $hDEF$ record nondegeneracy, which the classical picture leaves implicit. No auxiliary points or intermediate geometric constructions occur in `Proposition04.lean`.

8 Relationship with Earlier Propositions

8.1 Proposition I.2

The classical proof of SAS is expressed by Euclid through superposition rather than by a sequence of earlier construction propositions. Nevertheless, the preceding segment-transport theorem already establishes the general idea that congruent lengths can be reproduced independently of their original position.

In the Hilbert reconstruction, segment congruence is part of the foundational congruence structure, so no separate segment-copying construction is needed inside I.4.

8.2 Proposition I.3

Proposition I.3 develops the controlled cutting of one segment by a length congruent to another. It belongs to the same early congruence infrastructure, although it is not the decisive logical ingredient in SAS.

The contrast is useful: I.2 and I.3 are construction theorems about segments, whereas I.4 is the first major rigidity theorem for complete triangles.

8.3 Transition to Proposition I.5

Proposition I.5 uses SAS repeatedly. The auxiliary construction in the isosceles-triangle proof creates pairs of triangles with two corresponding sides and the included angle congruent, and I.4 propagates that local symmetry into further side and angle congruences.

Thus I.4 becomes one of the principal reusable congruence principles of Book I immediately after it is established.

9 Hilbert and Book Zero Components Used

9.1 Hilbert SAS

The decisive formal ingredient is the Hilbert SAS congruence axiom or its established interface theorem. Given two corresponding side congruences and congruence of the included angles, it supplies the remaining corresponding side and angle congruences.

The public Euclidean theorem therefore packages a foundational Hilbert congruence principle into the form of Proposition I.4.

9.2 Nondegenerate Triangles

The formal hypotheses explicitly require genuine triangles. The noncollinearity assumptions prevent the angle data from degenerating and make precise a condition that the classical diagram leaves implicit.

9.3 Angle Orientation

Angles are represented by ordered triples of points. Consequently the formal proof must ensure that the included angles supplied to SAS and the corresponding angles returned from it have exactly the orientations required by the Euclidean statement.

Symmetry and transport lemmas handle these changes explicitly.

9.4 Segment Congruence

The side hypotheses are expressed directly by the Hilbert segment congruence relation. No numerical length function or coordinate representation is introduced.

10 Formalization Notes

Proposition I.4 occupies a special position in the reconstruction.

For Euclid, SAS is proved by superposition: one triangle is imagined as being placed on another so that a side and the included angle coincide. The remaining equal side then forces the third vertex and hence the whole triangle to coincide.

In a Hilbert-style axiomatization, the same rigidity is foundational. SAS belongs to the congruence axioms. The Lean proof of the Euclidean proposition is therefore much shorter than Euclid's argument because the geometric content has already been placed in the Hilbert interface.

This is not a loss of information. It records a genuine difference in foundational organization:

$$\text{Euclid: superposition proof} \qquad \text{versus} \qquad \text{Hilbert: SAS congruence principle.}$$

The formal statement also makes nondegeneracy explicit. A diagram of two triangles silently suggests that each triple of vertices is noncollinear; Lean requires this information as part of the hypotheses needed to apply the angle-congruence machinery.

Proposition I.4 is therefore an early example where the Book I order and the logical order of the Hilbert foundation diverge sharply. What is a theorem for Euclid is already foundational infrastructure for the reconstruction.

11 Dependency Summary

```
Classical:
SAS data -> superposition -> coincidence -> I.4

Formal:
TriangleCongruentFromSAS -> euclid_proposition_4
```

No new local axiom is introduced by the Book I theorem. The proof burden has already been discharged in the Hilbert congruence layer.

12 Conclusion

Proposition I.4 is the first fundamental triangle-congruence theorem of Book I.

Its historical proof uses superposition, one of the most discussed features of Euclid's early geometry. In the Hilbert reconstruction the same mathematical rigidity is represented directly by the SAS congruence principle.

The resulting Lean theorem is therefore short for a substantive reason: the proof burden has moved from Book I into the axiomatic foundation.

This distinction becomes increasingly important in the propositions that follow. I.5 uses SAS as working infrastructure, and later constructions repeatedly rely on the ability to convert local side and angle congruences into congruence information about an entire triangle.

Proposition I.4 thus marks the point where the early segment constructions of I.1–I.3 give way to the systematic theory of triangle rigidity.